

NIST Digital Forensic Framework

- The digital forensic process widely adopted in investigations is based on guidance from the National Institute of Standards and Technology (NIST), particularly Special Publication 800-86.
- This framework provides a structured methodology to ensure investigations are systematic, repeatable, and legally defensible.
- It consists of four major phases: Collection, Examination, Analysis, and Reporting.

NIST Digital Forensic Framework

- **Collection Phase**

- The collection phase involves identifying, labeling, recording, and acquiring data from potential sources of digital evidence while preserving its integrity.
- Investigators must ensure that the evidence is not altered during acquisition. This is typically done using write blockers to prevent modification of storage media and by creating a bit-by-bit forensic image of the original device.
- Sources of evidence may include hard drives, RAM, USB devices, smartphones, cloud storage, and network logs. Every action taken during this phase must be thoroughly documented to preserve evidentiary value.
- In a real-life corporate data theft case, investigators may seize a suspect's laptop, disconnect it from the network to prevent remote tampering, and create a forensic image of the disk. Hash values are calculated before and after imaging to confirm that the copied image is identical to the original device.

QQ: What are write blockers? What is importance of imaging?

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- **Examination Phase**
- The examination phase involves processing the collected data to extract relevant information.
- During this phase, forensic analysts use specialized tools to recover deleted files, extract email communications, analyze browser histories, inspect registry entries, and examine file metadata.
- Techniques such as keyword searches, file carving, and timeline reconstruction are commonly employed to identify artifacts relevant to the investigation.
- For example, in a ransomware investigation, examiners may search for malicious executables, review system logs to identify the infection vector, and examine persistence mechanisms used by the malware.
- The objective is to isolate digital artifacts that contribute to understanding what occurred.

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- **Analysis Phase**
- The analysis phase focuses on interpreting the extracted data to reconstruct events and draw logical conclusions.
- Investigators correlate multiple data sources such as system logs, network traffic, file timestamps, and user account activity to determine the sequence of events.
- This phase transforms raw data into meaningful evidence by establishing timelines and identifying responsible parties.
- For instance, if logs show a user login at 10:05 AM, a USB device inserted at 10:10 AM, and confidential files copied at 10:12 AM, the analysis may conclude that the user performed unauthorized data exfiltration.
- The findings must be supported by documented evidence.

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- **Reporting Phase**
- The reporting phase involves documenting the entire investigative process and presenting findings in a clear, structured, and legally defensible format.
- The report must include the scope of the investigation, tools and methodologies used, evidence discovered, hash values, timelines, and expert conclusions.
- The language should be understandable to non-technical stakeholders such as attorneys or judges while maintaining technical accuracy.
- A well-prepared report ensures that another forensic examiner could replicate the investigation and reach similar conclusions.

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- **Chain of Custody**
- Chain of custody refers to the chronological documentation that records the handling, transfer, storage, and analysis of digital evidence from the moment it is collected until it is presented in court.
- This documentation establishes accountability and ensures that the evidence has not been altered or tampered with during the investigation process.
- A proper chain of custody record includes details such as the case identification number, description of the evidence, serial numbers, date and time of collection, the individual who collected it, transfer history, storage location, and signatures of all individuals who handled the evidence. Each transfer must be documented with precise timestamps.
- In a fraud prosecution case, if the defense argues that digital evidence was manipulated, the prosecution can present a complete chain of custody log showing every individual who handled the evidence and when.
- If the documentation is accurate and complete, the court is more likely to admit the evidence. However, if there are unexplained gaps in the record, the evidence may be deemed inadmissible.
- **QQ: Who is allowed to handle evidence in an investigation?**

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- **Evidence Integrity**
- Evidence integrity refers to maintaining the original state of digital evidence so that it remains authentic and unaltered throughout the investigative lifecycle.
- Preserving integrity is fundamental to ensuring admissibility in court and maintaining investigative credibility.
- Integrity is preserved by creating forensic images rather than analyzing original devices, using write blockers to prevent modification, storing evidence in tamper-evident bags, restricting access to authorized personnel, and calculating cryptographic hash values before and after analysis.
- **QQ: Why not analyze original data but the image instead?**

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- **Evidence Integrity**
- There are two primary acquisition approaches: static acquisition and live acquisition.
 - Static acquisition occurs when a device is powered off before imaging, minimizing the risk of data alteration.
 - Live acquisition is performed while a system is running, typically to capture volatile data such as RAM contents, active processes, and network connections.
- Although live acquisition may alter certain system artifacts, it is often necessary in incidents such as ransomware attacks where volatile memory contains critical encryption keys.
- In a malware investigation, forensic practitioners may first capture volatile memory to preserve running processes, then create a disk image and verify hash values. If the hashes remain consistent throughout the investigation, the integrity of the evidence is confirmed.
- **QQ: Why capture certain evidence live and others static?**

NIST Digital Forensic Framework

- **Hashing Basics**
- Hashing is the process of using a mathematical algorithm to convert data into a fixed-length string of characters known as a hash value or digest. This hash acts as a unique digital fingerprint of the data.
- Common hashing algorithms used in digital forensics include MD5, SHA-1, and SHA-256. Although MD5 and SHA-1 were widely used historically, they are now considered vulnerable to collisions.
- SHA-256, part of the Secure Hash Algorithm family developed under standards by the National Security Agency, is currently recommended because of its stronger resistance to collisions.
- Hashing plays a critical role in verifying evidence integrity. When a forensic image is created, the investigator computes a hash of both the original device and the forensic copy. If the two hash values match exactly, it confirms that the image is an exact duplicate. If they differ, it indicates potential tampering or corruption.
- **QQ: With algorithms like MD5 is it possible for someone to tamper with evidence? If yes, how?**

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- **Change Of Custody Mock Exercise**
- You are a digital forensic investigator assigned to investigate suspected insider data theft within a financial services company. Management reports that confidential client records were leaked.
- Suspicion falls on an employee in the accounting department. The employee's workstation and a USB flash drive found at their desk are to be seized as evidence.
- **Your task is to properly document and maintain the chain of custody from seizure to storage in the forensic lab.**

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- **Step 1: Identification of Evidence**
- **Step 2: Collection of Evidence**
- **Step 3: Initial Hash Documentation**
- **Step 4: Transfer to Forensic Laboratory**
- **Step 5: Secure Storage**
- **Step 6: Imaging and Return to Storage**

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Field	Details
Case ID	2026-FakeCase-001
Exhibit ID	A
Item Description	32GB SanDisk USB Flash Drive
Serial Number	SN-12345678
Date & Time Collected	04 March 2026 – 14:30
Collected By	Mr. X (Signed)
Collection Location	Accounting Office, Desk 4
Released By	Mr. X (15:15)
Received By	Ms. Y (15:20)
Purpose of Transfer	Transfer to Digital Forensics Lab
Storage Location	Evidence Locker #12
Date & Time Stored	04 March 2026 – 15:35

NIST Digital Forensic Framework

- **Search and Seizure in Digital Forensics**
- Search and seizure in digital forensics refers to the lawful process of identifying, collecting, and taking custody of digital devices or data for investigative purposes.
- Because digital devices often contain vast amounts of personal and unrelated information, search and seizure procedures must strictly comply with constitutional protections, statutory laws, and judicial authorization requirements.
- In criminal investigations, law enforcement officers typically require a valid search warrant issued by a court.
- A warrant must establish probable cause and clearly define the scope of the search, including the specific devices, locations, and types of data authorized for seizure.
- Overly broad or vague warrants may be challenged in court, potentially resulting in suppression of evidence. The warrant should specify whether investigators may search computers, mobile devices, cloud accounts, email records, or network logs.
- Digital searches differ from traditional physical searches because data may be hidden, encrypted, or stored remotely. As a result, courts often allow investigators to seize entire devices for off-site forensic examination.
- However, examiners must limit their analysis to data relevant to the scope defined in the warrant. Accessing unrelated files beyond authorized scope may violate privacy rights and render evidence inadmissible.

NIST Digital Forensic Framework

- **Admissibility of Digital Evidence**
- Admissibility refers to whether digital evidence is accepted by a court of law for consideration in legal proceedings. For digital evidence to be admissible, it must satisfy legal standards concerning **relevance, authenticity, reliability, and integrity**.
- Relevance means that the evidence must directly relate to the facts of the case and assist in proving or disproving a material issue. Courts will exclude evidence that does not meaningfully contribute to resolving the dispute.
- Authenticity requires proof that the evidence is what the investigator claims it to be. In digital forensics, authenticity is typically established through proper chain of custody documentation and cryptographic hashing.
- Reliability concerns whether the methods and tools used to obtain and analyze the evidence are scientifically valid and widely accepted.
- A digital forensic expert witness is a qualified professional who presents technical findings and opinions in court. The expert's role is to assist the judge or jury in understanding complex digital evidence that falls outside ordinary knowledge.

Real-Life Case: Ross Ulbricht- Silk Road Inves.

- **Case Background**
- Silk Road was an online black market operating on the Tor network (dark web).
- It facilitated the sale of illegal goods, especially narcotics.
- Payments were made primarily using Bitcoin to maintain pseudonymity.
- The site operated between 2011 and 2013.
- Ross Ulbricht allegedly operated the site under the alias “Dread Pirate Roberts.”
- The investigation was led primarily by the Federal Bureau of Investigation (FBI), alongside other federal agencies.

Real-Life Case: Ross Ulbricht- Silk Road Inves.

Investigation Techniques Used

- **Open-Source Intelligence (OSINT)**

- Investigators examined early online forum posts promoting Silk Road.
- Email addresses linked to early site promotion were traced back to Ulbricht.
- Username patterns were correlated across platforms.

- **Network Forensics**

- Analysis of IP address leaks from Silk Road servers.
- Identification of hosting infrastructure located overseas.
- Monitoring of traffic entering and exiting the Tor network.
- Correlation of server misconfigurations with real-world hosting providers.

- **Undercover Operations**

- Federal agents conducted undercover drug purchases on Silk Road.
- Transactions were traced through blockchain analysis.
- Internal Silk Road communications were gathered after server seizure.

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Real-Life Case: Ross Ulbricht- Silk Road Case.

- **Server Seizure and Digital Evidence**
- Authorities located and seized Silk Road servers hosted abroad.
- Forensic imaging (bit-by-bit copies) was performed on seized servers.
- Extracted data included:
 - User account records, Vendor information
 - Private messages, Bitcoin wallet data
 - Administrative logs
- Hash values were calculated to verify integrity of forensic images.
- **The Arrest (Live Digital Evidence Capture)**
- Ulbricht was arrested in October 2013 during a staged a distraction to prevent him from closing or encrypting his laptop.
- The laptop was seized while unlocked and logged in as site administrator.
- This enabled live acquisition of:
 - Active Silk Road admin panel, Private chats
 - Server control interfaces, Financial records
- This was a critical example of live digital evidence preservation.

Real-Life Case: Ross Ulbricht- Silk Road Case.

- **Chain of Custody**
- The laptop and digital storage devices were immediately documented.
- Investigators recorded:
 - Date and time of seizure
 - Location of arrest
 - Identity of collecting agents
 - Transfer to forensic labs
- Forensic imaging was documented step-by-step.
- Evidence was stored securely with restricted access.
- Proper documentation prevented admissibility challenges in court.

Real-Life Case: Ross Ulbricht- Silk Road Case.

- **Evidence Integrity and Hashing**

- Bit-by-bit forensic images were created from the laptop and servers.
- Cryptographic hashes (e.g., SHA family) were calculated before and after imaging.
- Matching hashes confirmed no alteration of evidence.
- Integrity verification was critical due to defense challenges.

- **Cryptocurrency Forensics**

- Bitcoin transactions were traced using blockchain analysis.
- Wallet addresses linked to Silk Road operations were identified.
- Financial trails connected Ulbricht to marketplace profits.
- Demonstrated that cryptocurrency transactions, while pseudonymous, are traceable.

- Ulbricht was convicted in 2015 on charges including:

- Narcotics trafficking conspiracy
- Money laundering conspiracy
- Computer hacking conspiracy

- The case became a landmark prosecution in cybercrime law enforcement

Documentation Standards And Reporting

- **Documentation Standards in Digital Forensics**
- Documentation in digital forensics refers to the systematic recording of every action taken during an investigation to ensure transparency, repeatability, and legal admissibility.
- Proper documentation allows another qualified examiner to reproduce the investigation and arrive at the same findings using the recorded procedures. It forms the backbone of credibility in both corporate and criminal investigations.
- **Forensic Reporting Structure**
- A forensic report is the formal written presentation of an investigator's findings. It translates complex technical processes into clear, structured, and legally defensible conclusions.
- A well-prepared forensic report must be objective, fact-based, and understandable to non-technical stakeholders such as judges, attorneys, corporate executives, or juries.

Documentation Standards And Reporting

- The report typically begins with an **executive summary**, which provides a concise overview of the case, objectives, and key findings. This section is written in plain language and highlights conclusions without overwhelming the reader with technical detail.
- The **methodology** section explains how the investigation was conducted. It includes the tools used, their versions, imaging procedures, hashing algorithms applied, and analytical techniques performed. This section ensures repeatability and demonstrates adherence to accepted forensic standards.
- The **findings** section presents factual results discovered during examination and analysis. It includes recovered files, log entries, timestamps, communication records, and relevant artifacts. Findings must be presented without speculation and supported by screenshots, hash values, or references to extracted data.
- The **analysis** section interprets the findings and reconstructs events based on evidence. Here, the investigator explains how data points correlate and what conclusions logically follow. Care must be taken to distinguish clearly between factual findings and professional opinions.

Ethics in Digital Forensic Investigations

- Ethics in digital forensic investigations refer to the professional principles and moral standards that guide investigators in conducting fair, impartial, and lawful examinations.
- One core ethical principle is **objectivity**. Investigators must remain neutral and avoid confirmation bias. Their role is not to prove guilt or innocence but to uncover and present facts based on evidence.
- **Confidentiality** is another critical ethical obligation. Investigators frequently encounter private communications, financial data, medical information, and trade secrets.
- **Integrity** and honesty are foundational ethical requirements. Investigators must never alter, fabricate, omit, or misrepresent evidence.
- **Competence** is also an ethical requirement. Digital forensic professionals must maintain up-to-date knowledge of tools, technologies, and methodologies.
- Legal compliance is another ethical dimension. Investigators must respect search and seizure laws, privacy regulations, and organizational policies.
- Finally, professional accountability ensures that forensic practitioners are prepared to defend their methods and conclusions under cross-examination.

NIST Framework for Digital Forensics

- What is a framework? Why NIST?
- What do you understand by a chain of custody broken?
- What are the legal parameters to be respected during a search and seizure of digital evidence?
- What is the role of an expert witness in a criminal / fraud case?
- Tor is secure and “impenetrable”... how in your opinion, did the FBI succeed to “penetrate” and analyze traffic from Silk Road Connected devices?
- What is the difference between anonymity and traceability?
- What are the key points crucial to a proper digital forensics report?
- Discuss the ethical points to be considered in digital forensics